JEEVAN-DROP - Low-Cost Flood Relief Water Filter

Pocket Water Filter System for North Bihar Flood Victims – An Inexpensive Design to Deal with Contamination of Ground Water

PROJECT REPORT

Industry Partner (Mentoring Purpose): Arosia Water **PROJECT GUIDE: Prof. Dr. Gunjan Kumar**

METHODOLOGY:

JEEVAN-DROP is a low-cost, zero energy water filtration system that can be deployed quickly in flood-hit areas such as North Bihar. This system runs purely on gravity, which means it can be easily installed on relief camp water taps or dispensers using a universal rubber connector. The system uses a three-level cartridge approach to filter highly turbid floodwater.





Fig 1 Prototype connected with tap for initial testing

1) **Prefiltration Stage:** The water initially passes through a cleanable polypropylene (PP) mesh pre-filter. It is a primary physical filter that removes mud, silt, and heavy sand, thus ensuring that the downstream fine membranes do not get clogged instantly.
2) **Chemical Adsorption Stage:** The second stage involves an activated carbon fibre block. It helps in removing any chemical pollutants, pesticides, and volatile organic compounds (VOCs). At the same time, it helps remove chlorine and foul odour to improve taste.
3) **Microbial Filtration Stage:** The last stage consists of a 0.2µm porous ceramic disc. It is a microscopic filter that physically filters out sediment, parasites (Giardia and Cryptosporidium), and over 99.9% of pathogenic bacteria like E. coli, cholera, and typhoid. The interlocking cartridges are housed in a compact, tool-free assembly initially developed using 3D-printed, food-safe PLA or PETG plastics for rapid prototyping.

PROPOSED BLOCK DIAGRAM:

Contaminated floodwater is first processed through a PP Mesh Pre-Filter to remove initial physical impurities. The flow then advances through an Activated Carbon Filter, followed by a 0.2-micron ceramic and maifan mineral ball stage for deep purification. This progressive filtration ultimately yields clean drinking water, delivering direct relief for affected people.

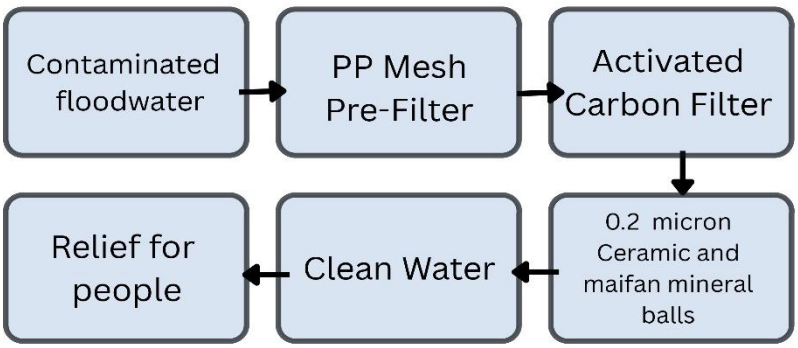


Fig 2 Block Diagram

PROPOSED OUTCOME:

The use of JEEVAN-DROP is to ensure quick and decentralized water security for displaced people without requiring any electricity or firewood. The results to be expected include:

1) **Disease Pathogens and Contaminants Removal:** Removes the pathogens and chemicals from a specific region using Maifan mineral balls for leaching minerals and creating ideal 6.5-8.5 pH balance.
2) **Incredibly Low Cost:** Ensuring the target retail cost of ₹150 to ₹170, making it nearly 90% cheaper than current commercial emergency filter straws.
3) **Sustainable Life-Cycle:** The modular structure of JEEVAN-DROP enables relief agencies or users to easily wash or replace the filter media without disposing of the whole device, guaranteeing zero plastic waste from the main housing. Overall, JEEVAN-DROP acts as a universal life-line, contributing to the achievement of the UN Sustainable Development Goal 6 (SDG 6).

In conclusion, the JEEVAN-DROP becomes an all-in-one lifesaver in helping to meet the United Nations Sustainable Development Goal 6 (SDG 6) (clean drinking water being a basic right).

Video Link: https://drive.google.com/file/d/1SQaBDgfYZ9FfooNY0X_eBunAM_gBPncU/view?usp=sharing